BUS 696 Generative AI in Finance
Course Syllabus
Spring 2026

Instructor: Jonathan Hersh, Ph.D
Associate Professor, Economics and Management Science
Argyros School of Business

Time: Wednesdays, 7:00pm - 9:50pm

Office: BK 307G

Office Hours: Wednesdays 6-7pm

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Course Description

Artificial intelligence is transforming finance—how markets process information, how trades are executed in microseconds, how risk is measured and managed, how ordinary people access financial advice once reserved for the wealthy. We are witnessing the emergence of what some call "AI-first finance," where algorithms don't just assist human decision-makers but increasingly replace them.

This is primarily a course in applied AI, meaning we will use the tools, lens, and framework of machine learning and data science to understand how finance is changing. We will read a lot in this course, but we will also run code—working with pre-built Jupyter notebooks to see how neural networks learn patterns in market data, how reinforcement learning agents discover trading strategies, and how backtesting reveals whether a model actually works or is just overfitting to noise. Topics we will cover include data-driven finance, neural networks, reinforcement learning, backtesting, risk management, algorithmic trading, cryptocurrency, and the future of AI in financial services.

Course Learning Outcomes

By the end of this course, you will be able to:

1. Explain fundamental financial concepts (returns, risk, portfolios) needed to understand AI applications
2. Describe how data-driven approaches have transformed finance
3. Understand key AI/ML techniques used in financial applications (neural networks, reinforcement learning)
4. Evaluate trading strategies through proper backtesting methodology
5. Assess risk using AI-enhanced approaches
6. Critically discuss the future trajectory of AI in finance and its implications

Readings

Textbook: *Artificial Intelligence in Finance* by Yves Hilpisch (O'Reilly, 2020)

Course Format

Class periods will consist of a mix of discussions and lectures.

Evaluation

Component	Weight
Participation & Lab Completion	20%
Problem Sets	25%
Research Paper Presentation – Present 20-25 minutes about a research paper with a * next to the title. Sign up here.	20%
Final Group Project & Presentation	35%

What success looks like in this course

This course emphasizes applied understanding over mathematical formalism. Strong performance means demonstrating that you can reason clearly about AI systems in financial contexts, interpret model outputs, and communicate tradeoffs and limitations to a business or policy audience. I care less about perfect code and more about whether you can explain what a model is doing, why it might fail, and whether it would be trusted in a real financial setting.

Use of AI tools

You are encouraged to use AI tools (e.g., ChatGPT, Claude, Copilot) as learning aids—for brainstorming, debugging, and improving clarity of explanation. However, you must understand and be able to explain any work you submit. Submitting AI-generated output that you cannot explain or defend will be treated as a failure to meet the assignment's learning objectives.

Course Resources

Website: I will use Canvas to distribute course materials.

Office hours**:** In general, I encourage you to come visit me during office hours!

Email: I will do my best to respond in a reasonable time. If I have not replied in 24 hours, please email again as it might have slipped my notice.

Poll Everywhere: We will on occasion use Poll Everywhere software. My polls can be reached via PollEv.com/hersh

Technology Policy–

My technology policy can be summarized as follows: ***I ask that you use technology thoughtfully in the classroom.*** Technology is wonderful, but the careless use of it can distract others and prevent you from being fully engaged with the material. Please completely silence your phones – *note, vibration mode is not silent* – and resist being distracted by texting, or viewing Facebook or other distracting website during class. **If I see that you are using technology for any purposes other than for course material, you will receive no points for class participation on that day.**

Classroom Etiquette–

Classroom is a sacred space. We should all strive to respect that space. This course will be hard at times, and that can be frustrating. I believe students learn best when they feel comfortable and can struggle with the material without fear of criticism from either the instructor or fellow students. Civility of peers and your instructor is expected at all times, both in the class and outside of it. Please come speak to me if you feel this is not being met.

Chapman University Academic Integrity Policy –

I take academic integrity very seriously. I am obligated to report any evidence of violations of Chapman's policy on academic integrity to the dean's office.

"Chapman University is a community of scholars that emphasizes the mutual responsibility of all members to seek knowledge honestly and in good faith. Students are responsible for doing their own work, and academic dishonesty of any kind will be subject to sanction by the instructor and referral to the university's Academic Integrity Committee, which may impose additional sanctions up to and including dismissal. (See the Undergraduate Catalog for the full policy.)

"Academic dishonesty can take a number of forms. It includes, but is not limited to, cheating on a test or examination, claiming the work of another as your own, plagiarizing any paper, research project or assignment (to appropriate for use as one's own passages or ideas from another), or falsely submitting material to fulfill course requirements."

Chapman Policy on Students with Disabilities –

In compliance with ADA guidelines, students who have any condition, either permanent or temporary, that might affect their ability to perform in this class are encouraged to inform the instructor at the beginning of the term. You also may contact the Office of Disability Services at www.chapman.edu/disabilities. The University, through the Disability Services Office, will work with the faculty member who is asked to provide the accommodations for a student in determining what accommodations are suitable based on the documentation and the individual student needs. The granting of any accommodation will not be retroactive and cannot jeopardize the academic standards or integrity of the course.

Important Addresses and Telephone Numbers

Disabilities Services:
410 N. Glassell
Phone: (714) 997-6778

Tutoring, Learning, and Testing Center:
Cecil B. DeMille Hall 130
Phone: (714) 997-6828

Chapman's Diversity Policy

Chapman University is committed to ensuring equality and valuing diversity. Students and professors are reminded to show respect at all times as outlined in Chapman's Harassment and Discrimination Policy. Please see the full description of this policy at http://www.chapman.edu/faculty-staff/human-resources/eoo.aspx. Any violations of this policy should be discussed with the professor, the dean of students and/or otherwise reported in accordance with this policy.

PART I: Introduction to Finance

Feb 4 – No class, travel to SF

Week 1 (Feb 11): Introduction: What is AI in Finance?

Topics:

- Course overview and expectations
- A brief history: from human traders to algorithms to AI
- Finance basics: What are stocks, bonds, and returns?
- Why AI is transforming finance now

Activity: Hands-on: Prompt engineering exercises with financial questions

Readings:

- Hilpisch, Chapter 1: Artificial Intelligence
- McKinsey, "The State of AI in 2025" (skim executive summary)

Week 2 (Feb 18): Finance Fundamentals for AI Applications

Topics:

- Understanding financial data: prices, returns, volume
- Risk and return: the fundamental tradeoff
- What is a portfolio? Diversification basics
- Key financial metrics: Sharpe ratio, volatility, drawdown

Activity: Lab: Exploring stock price data in Jupyter (calculating returns, visualizing prices)

Readings:

- Hilpisch, Chapter 4: Data-Driven Finance (pp. 81-100)
- Fama & French (1993), "Common Risk Factors in the Returns on Stocks and Bonds" — read introduction and conclusion only

Week 3 (Feb 25): Data-Driven Finance

Topics:

- How data transformed finance over the past 50 years
- Types of financial data: market data, fundamental data, alternative data
- The efficient market hypothesis and its challenges
- Introduction to quantitative approaches

Activity: Lab: Working with financial APIs (yfinance) to pull and analyze data

Readings:

- Hilpisch, Chapter 4: Data-Driven Finance (finish chapter)
- *Lopez-Lira & Tang (2023), "Can ChatGPT Forecast Stock Price Movements?"

📋 **Problem Set 1 assigned (due Week 5): Financial data exploration**

Week 4 (March 4): AI-First Finance: The New Paradigm

Topics:

- What does 'AI-first' mean in finance?
- Machine learning basics: supervised vs. unsupervised learning
- Neural networks: a conceptual overview (no math required)
- Where AI adds value in the financial services value chain

Activity: Demo: How a simple neural network makes predictions

Readings:

- Hilpisch, Chapter 6: AI-First Finance
- 2026 Anthropic Economic Index Report (focus on finance/workplace sections)
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PART II: AI Techniques in Finance

Week 5 (March 11): Dense Neural Networks for Market Prediction

Topics:

- Deep learning in finance: why it matters
- Dense neural networks (DNNs) explained simply
- Feature engineering: turning financial data into model inputs
- Overfitting: the #1 problem in financial ML

Activity: Lab: Running a pre-built DNN model on stock data, interpreting outputs

Readings:

- Hilpisch, Chapter 8: Dense Neural Networks (focus on concepts, skim code)
- López de Prado, "The 10 Reasons Most Machine Learning Funds Fail"

📋 **Problem Set 1 due**

Week 6 (March 18): Reinforcement Learning in Trading* Note class lecture will be delivered via asynchronous video – no in class meeting.

Topics:

- What is reinforcement learning? (agents, actions, rewards)
- RL vs. supervised learning: key differences
- Applications in trading: learning to trade from experience
- Challenges: simulation vs. real markets

Activity: Lab: Running a simple RL trading agent notebook — watch it learn and experiment with reward parameters

Readings:

- Hilpisch, Chapter 9: Reinforcement Learning

📋 **Problem Set 2 assigned (due week 8): Neural network experimentation**

March 25- No class, Spring Break

Week 7 (April 1): Backtesting: How Do We Know If It Works?

Topics:

- What is backtesting and why it matters
- The dangers: look-ahead bias, survivorship bias, overfitting
- Proper backtesting methodology
- Transaction costs and market impact

Activity: Student presentations: Case studies on trading strategy successes and failures

Readings:

- Hilpisch, Chapter 10: Vectorized Backtesting
- QuantStart, "Successful Backtesting of Algorithmic Trading Strategies" (online guide)

Week 8 (April 8): Risk Management with AI

Topics:

- What is financial risk? Types of risk in finance
- Traditional risk measures: VaR, Expected Shortfall
- AI-enhanced risk modeling
- Model risk: when your AI gets it wrong

Activity: Student presentations: Risk management case studies

Readings:

- Hilpisch, Chapter 11: Risk Management
- Coleman, "A Practical Guide to Risk Management" Chapter 1 (CFA Institute)

📋 **Problem Set 2 due**

PART III: Applications & The Future

Week 9 (April 15): Trading and Market Making* Note class will be delivered via asynchronous video due to conflict

Topics:

- Algorithmic trading: from simple rules to AI
- High-frequency trading basics and microwave networks
- Market making: providing liquidity
- The Flash Crash and market stability concerns

Readings:

- Kirilenko et al. (2017), "The Flash Crash" (skim)

📋 **Problem Set 3 assigned (due Week 11): Backtesting a simple strategy**

Week 10 (April 22): Bitcoin, Crypto, and DeFi

Topics:

- Blockchain technology fundamentals
- Bitcoin and cryptocurrency markets
- DeFi (Decentralized Finance): what is it?
- AI applications in crypto trading and analysis

Activity: Lab: Exploring crypto market data and volatility

Readings:

- Nakamoto (2008), "Bitcoin: A Peer-to-Peer Electronic Cash System"
- Shkilko & Sokolov (2020), "Every Cloud Has a Silver Lining: Fast Trading, Microwave Connectivity, and Trading Costs"*
- Harvey et al. (2021), "DeFi and the Future of Finance"

Week 11 (April 29): AI Agents, Robo-Advisors, and Consumer Finance

Topics:

- Robo-advisors: automated investing for everyone
- AI chatbots and customer service in finance
- AI agents: autonomous financial decision-making

- The future of financial advice

Activity: Student presentations: Fintech case studies (Wealthfront, Betterment, etc.)

Readings:

- Horton, "What Can We Learn from Homo Silicus?"*

📋 **Problem Set 3 due**

Week 12 (May 6): Alt Data and Market Intelligence

Topics:

- What is alternative data? (satellite imagery, social media, web scraping)
- Sentiment analysis and news analytics
- Generating financial reports with AI
- The alpha in alternative data: real or illusory?

Activity: Lab: Simple sentiment analysis on financial news

Readings:

- *Bartik, Gupta & Milo, "The Costs of Housing Regulation: Evidence from Generative Regulatory Measurement"
- Bollen, Mao & Zeng (2011), "Twitter Mood Predicts the Stock Market"

Week 13 (May 13): Final Presentations

Topics:

- Group project presentations (15-20 min each)
- Peer feedback and discussion
- Course wrap-up: What have we learned?
- Career paths in AI-driven finance

Activity: Final group presentations